

AI-Powered Research Literature Scout: Revolutionising Academic Research

Welcome to an innovative solution for academic research, designed to streamline the literature review process through the power of Artificial Intelligence and Machine Learning. This project introduces an automated system for fetching, ranking, clustering, and summarising research papers, transforming how students and researchers interact with vast amounts of information.

Automated Paper Fetching, Ranking, Clustering & Summarisation

Team 18: B.Tech 5th Semester



Addressing the Challenges of Literature Review

The Problem Statement

- Manual literature review is slow and inefficient
- Difficulty in filtering relevant research papers from extensive databases
- Lack of clear summaries and explicit connections between papers
- Students spend hours on irrelevant PDFs
- Teachers require structured summaries, not raw documents



Our Goal: An Automated, Intelligent System

- Searches research papers efficiently by topic
- Ranks papers using advanced ML embeddings
- Performs automatic clustering for thematic grouping
- Generates concise bullet-point relevance summaries
- Provides flexible export options (CSV / PDF)



Our Robust Technology Stack

Frontend

Built with HTML, CSS (Tailwind) for a modern, responsive UI, leveraging JavaScript and Server-Sent Events (SSE) for real-time progress updates. Features clean, animated interface with intuitive cards, sliders, and buttons.

Backend

Powered by FastAPI (Python) for high-performance API operations, featuring async processing (asyncio) and CORS middleware for seamless frontend communication.



Machine Learning

Integrates SentenceTransformer (all-MiniLM-L6-v2) for text embeddings, FLAN-T5 base for summary generation, MiniBatchKMeans for clustering, TF-IDF for keyword extraction, and Cosine Similarity for relevance scoring.

System Architecture: A Seamless Workflow



1. User Input

The user initiates the process by inputting a research topic.



2. Request to Backend

The frontend dispatches a POST request to the `/run` endpoint.



3. Async Job Execution

The backend commences an asynchronous job: fetching papers from arXiv API, computing embeddings, ranking by similarity, clustering, and generating summaries.



4. Real-Time Updates

The frontend listens via SSE to `/events/{task_id}`, displaying real-time progress, percentage completion, and live partial top papers.



5. Final Results

The comprehensive results are rendered as interactive cards, with options to export as PDF or CSV.

Key Features for Enhanced Research



Smart Research Paper Fetching

Utilises the arXiv API to retrieve cleaned titles, abstracts, authors, and publication dates.



Relevance Ranking

Converts paper abstracts and topics into embeddings, then applies cosine similarity for precise relevance scoring.



Automatic Clustering

Groups papers based on content similarity, automatically determining K clusters and extracting keywords using TF-IDF.



AI Summaries

Generates comprehensive cluster summaries and per-paper "Why relevant?" bullet points using FLAN-T5 for accuracy and conciseness.



Real-Time Progress UI

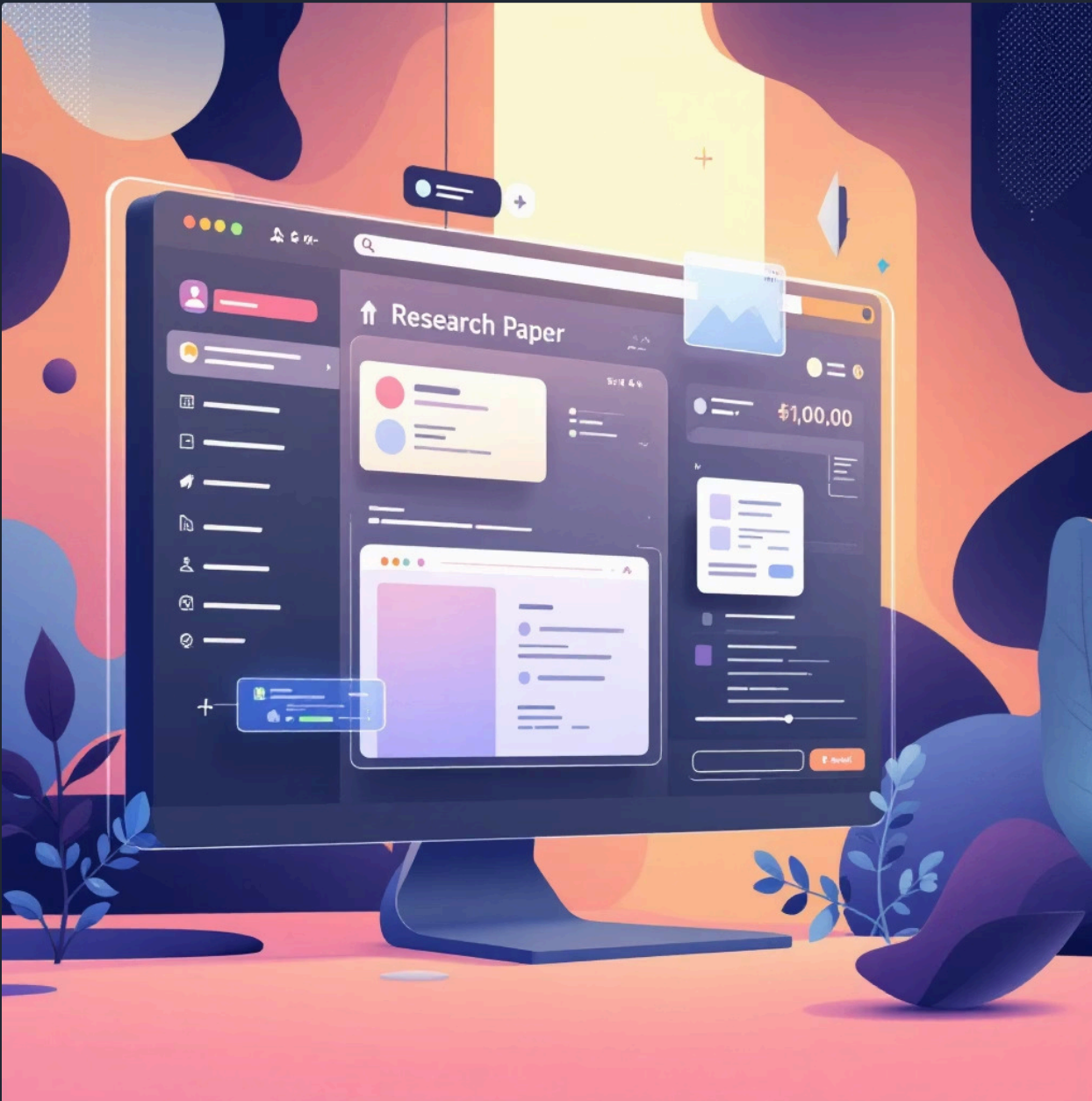
Provides live updates on stages (fetching, embedding, clustering, summarising) with a progress bar and ETA calculation.



Export Options

Offers flexible export capabilities for CSV and PDF formats (powered by ReportLab) for easy sharing and archival.

Intuitive User Interface (UI) Overview



Modern UI Elements

- TailwindCSS-based layout for aesthetic appeal
- Left sidebar for convenient controls and navigation
- Interactive sliders for paper count and top-N summaries
- Dynamic cards for each paper with expandable bullet points
- Scrollable results panel for efficient browsing
- Real-time, mid-screen progress bar for continuous feedback

Why This Matters

The UI is designed to be teacher-friendly, offering smooth interactions and a professional appearance, making demonstrations effortless.



Results and Future Outlook

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Outputs Generated

- Ranked papers with detailed relevance percentages
- Clearly defined cluster groups (2–6 clusters)
- Concise cluster summaries (90–120 words)
- Per-paper bullet points highlighting 3 key reasons for relevance
- Professional exports in both PDF and CSV formats

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Conclusion

This system automates 80% of the literature review workflow, integrating ML, NLP, clustering, and summarisation with real-time updates. It significantly saves time for students and researchers, demonstrating cutting-edge full-stack and AI integration.

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Future Improvements

- Integration with Google Scholar for broader scraping
- Advanced PDF parsing capabilities
- Implementation of user accounts for personalised experiences
- Support for locally running Large Language Models (LLMs)